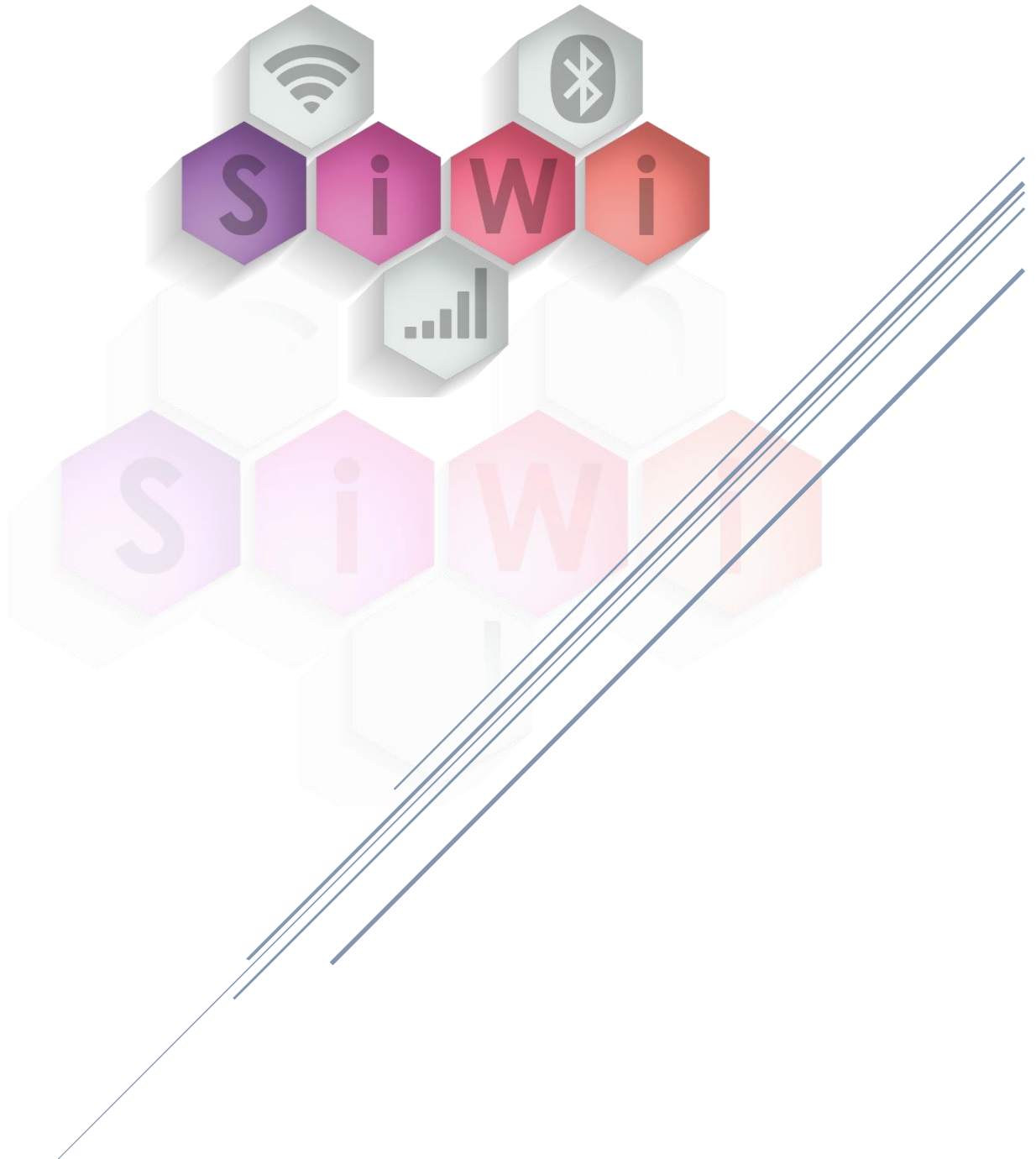


VTS PROTOCOL DESCRIPTION

Protocol Description for Chukast+ & Scout+ VTS Devices

v2.2



For Srishti Wireless Solutions

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Document History

Version	Date	Author	Description
1.0	2/22/2016	Ajay Bhargav (Waybyte)	Initial
1.1	12/1/2017	Ajay Bhargav	• Add section "Command Description" • Update protocol
1.2	12/19/2017	Ajay Bhargav	• Update PRD command
1.3	1/16/2018	Ajay Bhargav	• Update command section with biometric commands • Updated Transmit packet with sensor information
1.4	2/14/2018	Ajay Bhargav	• Update Time command as per latest firmware
1.5	2/5/2020	Ajay Bhargav	• Updated Protocol For new devices • Add new commands • Change command headings for easy reading
1.6	3/5/2020	Ajay Bhargav	• Update IMCONF command
1.7	8/12/2020	Ajay Bhargav	• Add TCP Protocol information • Add Transparent mode command • Add Aux UART mode command
1.8	9/8/2020	Ajay Bhargav	• Added Geofence Command • Update packet description for status flags
1.9	9/16/2020	Ajay Bhargav	• Update IMCONF command
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2.1	12/25/2020	Ajay Bhargav	• Add IMU Command • Update PRD command add Power save timer settings
2.2	12/24/2021	Ajay Bhargav	• Update IOATTACH command • Update Table 20, added more events to list

References

Sr. No.	Name	Author	Remarks

Abbreviations

Item	Description
JSON	JavaScript Object Notation
IoT	Internet of Things
SiWi	Srishti Wireless Solutions
APN	Access Point Name
SMS	Short Message Service
LUA	The programming Language (https://www.lua.org/about.html)

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Chaukas+ & Scout+ for Tracking Application

SIWI VTS supports its legacy mode of communication over TCP socket and simple to parse CSV packet. There are no complicated login procedures to follow or no need to send heartbeat packets to keep track of server connectivity. Device works on its own with TCP level acknowledgement to maintain connectivity and ensuring zero packet loss.

Keeping the latest trends of market in mind we have introduced new data transfer method. Now instead of sending data to server via traditional methods like TCP/UDP, device can send data directly to your HTTP server or web API and in an extremely easy to parse format JSON. Since JSON is one of the common data exchange method available in all web technologies, so implementing a receiving frontend for SiWi VTS Products will be much easier than writing a TCP listener. This will also help lower down your web server requirements and hence the overall cost of implementation.

About JSON

JSON (JavaScript Object Notation) is a lightweight data-interchange format. It is easy for humans to read and write. It is easy for machines to parse and generate JSON data. JSON is a text format that is completely language independent but uses conventions that are familiar to programmers of the C-family of languages, including C, C++, C#, Java, JavaScript, Perl, Python, and many others. These properties make JSON an ideal data-interchange language.

Why JSON?

The reason for introducing JSON to VTS was to ease and expedite front end development for it. Since most of the web developers are very much familiar with formats like JSON and ample amount of support available on web. Moreover, all web development platforms support JSON and its parsing. This is the reason JSON has become common data exchange format between IoT devices and web services.

JSON Transmit Protocol Description

Sample JSON Packet for Locator sent to server via HTTP/HTTPS POST method:

```
{
  "uid": 222,
  "info": {
    "dt": 1512141732,
    "txn": "E",
    "msgkey": 0,
    "msgid": 11720,
    "cmdkey": ""
    "cmdval": ""
  },
  "gps": {
    "fix": "A",
    "loc": [22.061200, 74.133017],
    "speed": 0,
    "sat": 10,
    "alt": 14,
    "dir": 137,
    "odo": 1332893
  },
  "io": {
    "box": 0,
    "ign": 1,
    "gpi": 1,
    "status": 0,
    "analog": [1935, 1234]
  },
  "pwr": {
    "main": 1,
    "batt": 1,
    "volt": 3826,
    "mvolt": 12.36,
  },
  "dbg": {
    "status": [5, 5, 22, 8, 3, 1],
    "ver": ["X.YY", "A.BB"],
    "lib": "6.65"
  }
}
```

NOTE: It is not possible to include all sample packets based on different sensor combinations, hence only basic device packet is mentioned here. Please cross check the actual packet with field description table provided below.

Table 1: Transmit Protocol Field Descriptions

Key	Data Type	Description	Value
uid	Number	Unit/Device ID (Software selectable, can be a numeric ID or IMEI of device)	
info	Object	Packet related info, Refer Table 2: Information Object	
gps	Object	GPS related data, Refer Table 3: GPS Data Object	
io	Object	Device's Digital and Analog IO information, Refer Table 4: Device IO Object	
pwr	Object	Device power parameters, Refer Table 5: Power Status Object	
dbg	Object	Device debug parameters, Refer Table 6: Debug Status Object	

Table 2: Information Object

Key	Data Type	Description	Value
dt	Number	Date time represented in Unix Timestamp format (Seconds since 1 Jan, 1970)	1512141732
txn	String	Transmission Reason, A packet identifier. Refer Section Packet Identification for list of packet Identifiers	"E"
msgkey	Number	This field provide information regarding key number during key press event in the device, currently kept as reserved	0
msgid	Number	A numeric counter for data packet.	11720
cmdkey	String	Command key identifier	"prd"
cmdval	String	This field contains value of the parameter when changed or requested from server or SMS or command line interface	"60"

Table 3: GPS Data Object

Key	Data Type	Description	Value
fix	String	GPS fix status A – For Valid Data V – For invalid data	"A"
loc	Number Array	Location represented as array of Number [Latitude, Longitude]. Both Latitude and Longitude are represented as decimal number up to a precision of 10 and a scale of 7.	[22.061200, 74.133017]
speed	Number	GPS speed parameter in Km/h	0
sat	Number	GPS Satellites in view	10
alt	Number	GPS altitude in meters	14
dir	Number	GPS direction/Bearing in degree	137
odo	Number	Device Odometer in meters	1332893

Table 4: Device IO Object

Key	Data Type	Description	Value
box	Number	Device enclosure open/close status 1 – Open 0 – Close	0
ign	Number	Vehicle ignition status 1 – Ignition On 0 – Ignition Off	1
gpi	Number	General purpose inputs represented in binary format e.g., A value of 12 represents in binary as 0001100 which means GPIO, GP1, GPI4, GPI5, GPI6 – Off GP2, GP3 – On	0
status	Number	Status Flags: Bitwise status information see Table 11: System Status Flags for more details.	0
analog	Number Array	Analog input value represented in millivolts	[1234,1234]
fuel¹	Object	Fuel Data vary based on the fuel source selected	
sensor²	Object Array	Sensor data object. Refer Table 9: Sensor Data object for more information	

Table 5: Power Status Object

Key	Data Type	Description	Value
main	Number	Power mains input availability status 1 – Mains Available 0 – Mains Unavailable	1
batt	Number	Battery connections status 1 – Battery Connected 0 – Battery Unconnected	1
volt	Number	Battery Voltage in millivolts	3826
mvolt	Number	Main battery input voltage in volts	12.36

Table 6: Debug Status Object

Key	Data Type	Description	Value
status	Number Array	Debug array: [CREG, CGREG, CSQ, Net, Server, Offline] CREG – GSM Registration status CGREG – GPRS Registration Status CSQ – Signal Quality Net – Network debug field Server – Server connectivity debug field (Table 7: System's Server State) Offline – Packet is online or offline indicator (1 – offline packet/retransmitted, 0 – Online packet)	[5, 5, 22, 8, 3, 1]

¹ Fuel key is not available in all protocols, devices with Fuel sensor installed will have this key present.

² Sensor key is not available in all protocols, devices with temperature sensor installed will have this key present.

ver	String Array	Device Application software version and hardware version	["1.9JCPLUS", "1.0"]
lib	String	Device firmware library version	6.65

Table 7: System's Server State

Index	Description
0	Invalid or unknown state
1	No network registered
2	GSM network registered
3	GPRS & GSM network registered
4	Trying to connect to programmed server IP & Port
5	Connected to programmed server IP & Port
6	Currently sending data to server

Table 8: Fuel Sensor Data Object

Type	Key	Data Type	Description	Value
Analog/MODBUS/UART Type Sensor	level	Number	Fuel Level in mV, to be calculated on server	1234
	err	Number	Error code if sensor fails	0
Omnicom Digital Sensor	f	Number	Device sends all sensor parameter as it is to server, please refer Omnicomm sensor datasheet for more information.	
	t	Number		
	n	Number		
	err	Number	Error code if sensor reading fail	
Ultrasonic Sensor UFS-1/UFS-2	type	String	Device sends sensor information to server as it is. Please read sensor datasheet for more information	
	llv	Number		
	rtv ³	Number		
	temp ⁴	Number		
	tilt ⁴	Number		
	cvl ⁴			
	sig	Number		
	swc	String		
	hwc	String		
	err	Number	Error code if sensor reading fail	

³ UFS-1 Type Sensor only

⁴ UFS-2 Type Sensor only

Table 9: Sensor Data object

Type	Key	Data Type	Description	Value
Temperature Sensor	temp	Number	Temperature value is represented as temp * 100. E.g. if value is 1234 then temperature is 12.34 °C	1234
	alarm	Number	Alarm bits for sensor: 0 – No Alarm 1 – Temperature Low Alarm 2 – Temperature High Alarm	0
	err	Boolean	True If sensor error, false if no error	False
Temperature & Humidity Sensor	temp	Number	Temperature value is represented as temp * 100. E.g. if value is 1234 then temperature is 12.34 °C	1234
	humid	Number	Humidity value is represented as humid * 100. E.g. if value is 1234 then temperature is 12.34%	1234
	alarm	Number	Alarm Bits for sensor Encoded as Shown in Table 10: Temperature & Humidity Error Bit Encoded	
	err	Number	Error code if sensor read fail, 0 for no error	

Table 10: Temperature & Humidity Error Bit Encoded

BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0	0	Humidity High Alarm	Humidity Low Alarm	0	0	Temperature High Alarm	Temperature Low Alarm

E.g., If Sensor has Temperature Low and Humidity High alarm active then alarm value will can be computed as shown below:

BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0	Alarm Value
0	0	1	0	0	0	0	1	33 (Decimal) 21 (Hex)

Table 11: System Status Flags

BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
NA	NA	NA	NA	Trip Status	Geofence Status	Immobilizer Status	Box Status

TCP Transmit Protocol Description

TCP Packet for Locator sent to server over TCP socket in a CSV (comma separated value) format. The sample packet is shown below:

\$SIWI,1234567890123,45,E,,89917650642221590319,0,1,0,12.36,141313,0,9,A,21.981985,85.221165,804,280,105113,220420,17,1,13,5,4071,1,0,8,0,0,0,1.0,1.2CPLUS_6.52,0!

Table 12: Transmit Protocol Field Description

#	Field Name	Description	Example
1	Header	Fixed Device Packet header “\$SIWI”	\$SIWI
2	Unit ID/IMEI	Unique Unit ID of device, this field is configurable to show Unit ID or IMEI	1234567890123
3	Message ID	Message ID of packet after power on. This field resets to 1 when device reboots.	45
4	Transmission Reason	Transmission Reason, A packet identifier. Refer Packet Identification for list of packet Identifiers	E
5	Command Key	Command identifier, command sent to device is set in this field during device response to server.	
6	Command Key Value	This field contains value of the parameter when changed or requested from server or SMS or command line interface. This field also carries SIM ID when device connects to server for the very first time after power on. SIM ID can be useful for identifying which SIM card is installed in the device	89917650642221590319
7	Ignition	Ignition Status, 0 – Ignition Off, 1 – Ignition On	0
8	Power Cut	External Power connection status, 0 – External Power disconnected, 1 – External Power Connected	1
9	Status Flags	status flags representing status of immobilizer, geofence, trip etc. in bit wise field represented as decimal value. See Table 15: System Status Flags for more details	0
10	Mains Voltage	Mains voltage in volts (float)	12.36
11	Odometer	Odometer value in meters	141313
12	Speed	Vehicle speed in Km/h	0
13	Satellite in View	GPS + GLONASS Satellite used for position acquisition	9
14	GPS Data Validity	A – Fixed Data V – Void or Invalid Data	A
15	Latitude	GPS Latitude Value in degrees	21.981985
16	Longitude	GPS Longitude Value in degrees	85.221165
17	Altitude	Altitude in meters	804
18	Direction	Direction or Course over ground in degrees	280
19	Time	Time in HHMMSS format. Timezone can be configured in device, default timezone is UTC.	105113
20	Date	Date in DDMMYY format.	220420

21	Signal Strength	GSM Signal strength value	17
22	GSM Status	GSM Network registration status	1
23	Error Code / Miscellaneous Field	Field used for error codes in temperature devices or use for other purpose based on specific firmware. Refer Table 14: Sensor Error Bit Encoded for more information	
24	Server Status	Server connectivity debug field (Table 13: System's Server State)	5
25	Internal Battery	Internal battery voltage in millivolts. Divide by 1000 to get value in volts.	4071
26	Analog Input	External Analog input value in millivolts/10. Divide by 100 to get value in volts	123
27	Digital Input Status	General purpose inputs represented in binary format e.g., A value of 12 represents in binary as 0001100 which means GPIO, GP1, GPI4, GPI5, GPI6 – Off GP2, GP3 – On	0
28	Aux Field 1	Auxiliary field 1 carry information based on type of sensor connected to device and 0 otherwise.	
29	Aux Field 2	Auxiliary field 2 carry information based on type of sensor connected to device and 0 otherwise.	
30	Aux Field 3	Auxiliary field 3 carry information based on type of sensor connected to device and 0 otherwise.	
31	Aux Field 4	Auxiliary field 1 carry information based on type of sensor connected to device and 0 otherwise.	
32	Hardware Version	Device hardware design version	1.0
33	Software Version	Device firmware version and Library version used by firmware, separated by “_” (underscore)	1.9CPLUS_6.65
34	Packet Type	Packet is online or offline indicator (1 – offline packet/retransmitted, 0 – Online packet)	0
35	End Delimiter	End marker “!<CR><LF>”, You can use “!” To split multiple packets if transmitted together by ISP relay server.	!

Table 13: System's Server State

Index	Description
0	Invalid or unknown state
1	No network registered
2	GSM network registered
3	GPRS & GSM network registered
4	Trying to connect to programmed server IP & Port
5	Connected to programmed server IP & Port
6	Currently sending data to server

Table 14: Sensor Error Bit Encoded

BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
NA	NA	NA	NA	Sensor4	Sensor 3	Sensor2	Sensor1

NA - Fields in the table are not used and always read as 0.

BIT [0:4] – Each bit is assigned to a sensor. If corresponding bit is 1 that sensor is faulty or not responding.

E.g. If Sensor 3 and Sensor 1 are at fault the error value will can be computed as shown:

BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0	Error
0	0	0	0	0	1	0	1	5

Table 15: System Status Flags

BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
NA	NA	NA	NA	Trip Status	Geofence Status	Immobilizer Status	Box Status

Packet Identification

There are various events and normal data which can be generated by device at runtime e.g., when someone disconnects the main power or when ignition status is changed etc. These events are sent to server in JSON format with a unique packet identifier for server application to take decision based on type of event occurred. Packet identified is sent with a key name of “**txn**” in information object of JSON data (**Table 2**). When device connects to server for the first time after reboot or a reset, command value “cmdval” will carry SIM card serial number (ICCID).

There are two types of Packet identifiers possible in the data packet

- A. Periodic Packet ID
- B. Event Packet ID

Periodic Packet ID

Device sends data to server periodically at an interval set by user. Periodic data ensures device status gets updated to server at a regular interval. Following is a list of periodic packet IDs:

Table 16: Periodic Packet Identifier

Identifier	Description
A	Vehicle in Normal Motion: This packet ID is sent during periodic transfer when vehicle is in motion.
B	Vehicle in motion but over speeding: This packet ID is sent during periodic transfer when vehicle is moving at higher speed that set limit.
E	Vehicle not in motion: This periodic event is generated when device is at halt or not moving.

Event Packet ID

Event packets are generated when device sense a change in expected status from the set value. Following is a list of event packet identifiers:

Table 17: Event Packet Identifier

Identifier	Description
C	Vehicle Speed Normal: This is an event packet ID and generated when vehicle lowers its speed than set limit. After this event, periodic event ‘A’ will be sent since device is in motion
D	Vehicle speed limit exceeded: This event is generated when vehicle crosses its set speed limit. After this event, Periodic event ‘B’ will be sent.
F	Vehicle Motion start: This event is generated when vehicle starts to move. After this event periodic event ‘A’ is generated.
G	Vehicle Motion Stop: This event is generated when vehicle stops or speed goes to 0. After this event periodic event ‘E’ is generated.
H	GSM Location Used: This event is generated when instead of GPS, GSM triangulated location is used – Only for portable devices
	Tilt Alert: For devices with MEMS sensor installed.
I	GPS First Fix: This event is generated when first fix of GPS is received. This packet is only generated when GPS data is good enough to considered as fixed after data post processing.

J	Ignition Change: This event is generated when ignition status changes from either on to off or off to on. The status of ignition will be updated in the IO structure.
K	Box/Enclosure Open: This event is generated when device is opened by someone. This can help preventing any tempering with the device. The status of box will be updated in the IO structure.
L	Power Transit Event: This event is generated when external power/main supply is connected/disconnected. The power status field will be updated with status of mains power.
M	Harsh Turn Detected: This event is generated when vehicle makes a sharp turn.
N	SOS Event: This event is generated when SOS key is pressed on device
O	Digital IO/GPI changed: This event is generated when any of the general-purpose input pin see a transition. The GPI filed in IO structure is updated.
P	Vehicle Over Acceleration: This event is generated when vehicle acceleration is more than a set rate
Q	Parameter Change Event: This event is generated when device parameter is changed. The “cmdval” field in info structure is populated with the changed parameter value.
R	Parameter Query Event: This event is generated when a query command for parameter is sent to device. The “cmdval” field in info structure is populated with the parameter value.
S	Harsh break Event: This event is generated when vehicle experience a harsh break.
T	Immobilize Status Event: This event indicates immobilizer enable/disable. “cmdval” field is populated with status of immobilizer.
U	GPS Status Event: This event is generated when status of GPS changes. “cmdval” defines the status of GPS. 0 – GPS Not Responding 1 – GPS OK 2 – GPS Lost 3 – GPS Acquired
V	Temperature/Temperature & Humidity Alarm Alert: This event is generated to provide high or low alerts for temperature/humidity, only available in devices with respective sensor installed.
W	New ID Alert: This event is generated only when new ID is detected, cmdval will have the value of detected ID. ID value can be Card ID for RFID card, Fingerprint ID for Biometric sensor or Chip ID for iButton Key.
X	Geofence Alert: This event is generated when vehicle enter/exit a geofence. cmdkey will be “1” For geofence entry and “0” for geofence exit cmdval will have geofence name set during geofence configuration.
Y	Trip Alert: This event is generated when trip button is pressed to detect start/end of trip. Only available in devices with Trip switch.
Z	FOTA Event: This event is generated when device firmware upgrade is started

JSON Receive Protocol Description

When device POST data to web server it expects server to respond with JSON data in following format:

Response with command:

```
{
  "result": true,
  "msg": "Data Success",
  "cmd": "prd=30"
}
```

Response without command (Use this if there is no command to be sent to device):

```
{
  "result": true,
  "msg": "Data Success",
}
```

Table 18: Receive Protocol Field Description

Key	Data Type	Description	Value
result	Boolean	Represents if data is accepted by server or not	True
msg	String	Message from server, can be null	"Data Success"
cmd	String	This field may or may not exist. If server has any command to be sent to device, this field can be populated else do not include in the response.	"prd=120"

TCP Receive Protocol Description

Device does not expect any kind of response from server. However, server can send commands to device for configuration or query any parameter of the device. The format of command from server to Device is same as that of SMS except there is no need to provide SMS password.

e.g., For configuration in via SMS command might be:

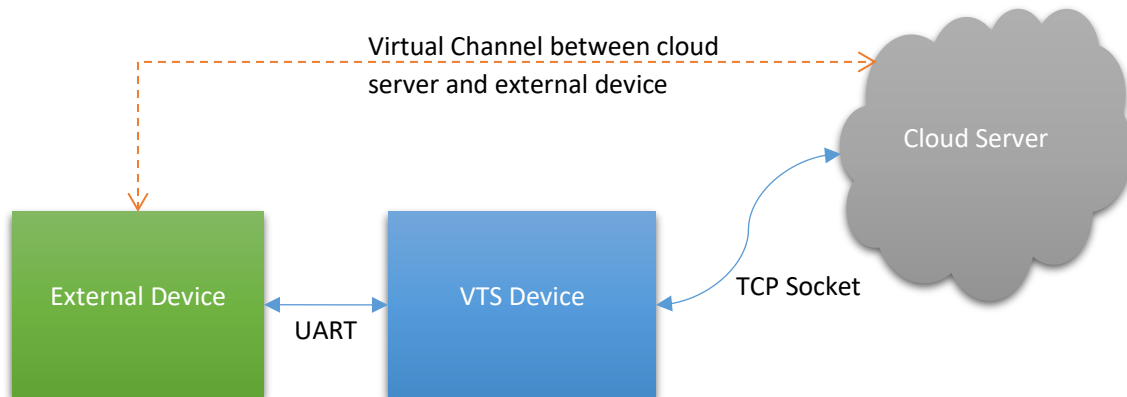
ttysms,PRD=60

In case of TCP, server can directly send:

PRD=60

Further section will cover commands in detail.

Transparent Mode for Auxiliary UART



VTS has auxiliary port for communication with external devices with firmware support available for many different types of devices e.g., RFID, Fuel, UHF reader etc. However, there are some use case where user need to get direct access of the device remotely. This is where transparent port comes into play.

In transparent mode, VTS device open a secondary TCP connection and sends its identification to the server in format shown:

```
$TPUART,[Device UID],[Device IMEI]!
```

This helps server maintain individual connections based on device UID or IMEI.

Once connection is established, TCP connection works as a transparent channel between Auxiliary UART port and server. Whatever is sent from server is directly transferred to UART and data from UART port is sent directly to server.

Transparent mode can be helpful in managing external device as well as to debug external device in case of failure.

To set UART into transparent mode user can send **UMODE** – Aux UART Operation Mode command.

Transparent mode parameters like server IP, server port and UART communication settings can be done using **TPCFG** – **Transparent mode configuration** command.

Command Description

Device supports configuration that can be done from either server or via SMS. Commands sent over SMS are protected via SMS password. User can change this password anytime if they feel security is compromised in anyway.

Commands sent from TCP are in simple format as shown:

cmd=parameters

Commands sent via SMS are sent with passwords as shown:

password,cmd=parameters

Default password is “tysms”. So, if user has not changed any password then, commands can be sent as:

tysms,cmd=parameters

SiWi Modbus device support two types of commands:

- A. Configuration Commands: To configure device over the air using TCP/SMS. All configuration commands require SMS password when sending over SMS.
- B. Service Commands: Service level commands used during installation and troubleshooting. All commands can be sent over SMS, and not all commands are allowed over TCP. The one which are allowed will be mentioned separately.

Configuration Commands

1. TIME – Configure Clock

Set/get/sync system clock. VTS device can sync its clock from three different time sources, following is the list based on source priority:

1. NSP – Network Service Provider (Lowest priority)
2. NTP – Network Time Protocol
3. GPS – GPS clock (highest priority)

Usage:

time - Get current operating mode and system date & time

time=[timestamp] - Set system time, where timestamp is Unix Timestamp/EPOCH Time

time=auto – Set Auto synchronization mode with time sources

time=manual – Disable Auto mode, User can set time using timestamp

time=mode – Get Current operating mode (Auto/Manual)

time=server – Get current NTP server

time=server,[NTP Server address] – Set NTP server address, default is siwi.in

time=src – Get currently used time source.

2. SMS – Admin Number Setting

Set SMS or call verification numbers. Currently only used for devices with call facility. SMS are currently protected by SMS Password only.

Usage:

sms=<index>,[<value>]

Index – Index to read/write cell number (1-5)

value – Cell number to be stored, use value as “Delete” to remove stored number

3. NIP – Server IP/URL

Set server URL for posting JSON data or Server IP or domain for TCP data.

Usage for TCP:

nip=example.com

Usage for JSON:

nip=http://xxx.xxx.xxx.xxx/post.php

nip=http://www.example.com/post.php

if use is using a different port then use as shown

nip=http://www.example.com:port/post.php

4. NPT – Set Server Port

This command is used in case of TCP mode only and is used to set server port.

Usage:

npt=8446

5. PWD – Change SMS Password

Set Device SMS password

Usage:

pwd=<new>

new: New password

Note Default password is “ttsms”.

6. REBOOT – Device reboot

Reset or Reboot device

Usage:

reboot – No parameters needed

reboot=<delay>

Where delay is number of seconds to wait before reboot.

7. UNO – Protocol UID Configuration

Select to use Device ID (UID), IMEI or custom name (UNAME) of the device in UID filed of protocol.

Usage:

uno=[imei/uid/uname]

8. UNAME – Device Custom Name

Set User defined name for device

Usage:

uname=<value> - where value is new device name in alphanumeric format (50 characters max)

9. OFR – Clear Offline Data

Clean and reset offline data

Usage:

ofr – Get currently stored offline packet count

ofr=1 – Clear Offline storage

10. DNSCFG – DNS Server Configuration

Configure custom DNS servers. By default, device will use DNS server provided by GSM service provider.

Usage:

dnscfg=primary,secondary

dnscfg=reset – Reset DNS server to default (reboot required)

e.g.

dnscfg=8.8.8.8,8.8.4.4

11. TZN⁵ - Timezone Configuration

Device Time zone setup

Usage:

tzn=<+/-HH:MM>

12. APN – Access Point Configuration

Set APN information for internet access

Usage:

apn=apn_name,[apn_username],[apn_password]

apn_name: Access point name. Can be set to "auto" for auto APN feature.

apn_username: APN username (Optional)

apn_password: APN user password (Optional)

13. PRD – Data Period Configuration

Set data transmission period/rate

Usage (For Non-Portable Devices):

To set all timings at once:

prd=<value> - where value is time in seconds

To set ignition dependent data transmission rate.

⁵ TZN will not affect UNIX timestamp since they are always represented in GMT.

For Ignition off transmission rate:
prd=ioff,<value>

For Ignition On transmission rate:
prd=ion,<value>

To set Ignition independent data transmission rates:

For transmission rate when vehicle is moving:
prd=move,<value>

For transmission rate when vehicle is stopped:
prd=stop,<value>

NOTE: You can either set Ignition dependent or independent transmission rates

Usage (For Portable Devices):

prd=<value> - where value is time in seconds

To set wakeup time in Power save mode (applies to both Normal and Portable devices):

prd=psave,<value> - Where value is time in seconds. Default 3600 (60 mins).

14. WGET – Download File

Download file on device via HTTP/FTP

Usage:

wget=<localname>,<http or ftp url>

e.g., for HTTP:

wget=reginfo,http://www.siwi.in/samplereg.txt

e.g.: for FTP:

wget=reginfo,ftp://username:password@siwi.in/public_html/samplereg.txt

15. DFMT – Date Format Setting

Set date time format

VTS device now support two different date time formats in JSON packet.

- A. EPOCH
- B. String (DDMMYYHHMMSS)

Default date time format in JSON packet is EPOCH/Unix timestamp.

Usage:

DFMT=EPOCH – Set date time format to EPOCH

DFMT=String – Set date time format to DDMMYYHHMMSS

16. ODM – Initial Odometer configuration

Set initial odometer value. It is sometimes necessary to set a start value of odometer during new installation if user do not want odometer to start from 0. This command sets the initial value of odometer.

Usage:

odm=<value>

Value of odometer is in meters, e.g. if user wants to set 1950Km as initial value then command should be odm=1950000

17. ESS – Speed limit Setting

Set excess speed value or maximum speed limit of vehicle. After the maximum speed limit of vehicle is set server will be notified if device moves at a higher speed than set value. If buzzer is installed in the device, then buzzer will also beep locally to notify driver of speed limit reached.

Usage:

ess=<value>

Value is in Km/h

18. STARTV/STOPV – Immobilizer Commands

These are immobilizer commands and only available in device with immobilizer feature. These commands do not have any special parameters.

Startv – Start the vehicle

Stopv – Stop the vehicle

19. HBRK – Harsh Break Threshold

Set harsh break threshold in g ($1g = 9.8m/s^2$). Default value is 0.55g

Usage:

hbrk=<value>

20. HACC – Harsh Acceleration Threshold

Set harsh acceleration threshold in g ($1g = 9.8m/s^2$). Default value is 0.43g

Usage:

hacc=<value>

21. EVSET – Event Configuration

To enable or disable event packets.

Usage:

evset=<list of txn to disable>,[0/1]

TXN list can be found in **Table 17: Event Packet Identifier**

22. SOS – SOS Acknowledge

Server command to acknowledge SOS packet. This will provide LED indication to person in vehicle that server has acknowledged SOS event sent by device. This command has no parameters.

23. TEMPTH⁶ - Temperature Alert Setting

Set temperature threshold.

Usage:

tempth=<id>,<high value>,<low value>

Where high and low value are the temperature thresholds in °C

24. RHTH – Humidity Alert Setting

This command is used to set humidity sensor thresholds in percent. Default values are 5% for low and 80% for high.

Usage:

RHTH=<ID>,<HIGH>,<LOW>

Where ID is sensor ID.

High and low are relative humidity thresholds in percent.

25. SENSOR – View/Set sensor information

This command is used to view real time sensor values and set per sensor offsets. For temperature sensor it is also used to reset the sensor if needed during installation.

Usage:

To Reset Temperature sensor

SENSOR=[ID],reset

To View current offset values:

SENSOR=[ID],offset

To Configure Temperature sensor offset

SENSOR=[ID],[Temp_offset]

To Configure Temperature and Humidity sensor offsets

SENSOR=[ID],[Temp_offset],[Humid_offset]

ID is sensor ID starting from 1.

To Set Temperature sensor count installed on device:

SENSOR=temp,<count>

To Set Temperature & Humidity sensor count installed on device:

SENSOR=trh,<count>

Temperature and Humidity offset values are in float can be negative or positive.

NOTE: Please be careful when resetting the sensor. Only use it when really required. Contact support team for any help.

⁶ This command is only available in devices with temperature sensors installed.

26. BUZZ⁷ - Buzzer Control

To turn buzzer on or off from server

Usage:

buzz=<duration>

Duration is provided in seconds, to turn off duration set as 0.

Buzz=<type>,<on/off>

Type can be:

SL: Speed Alert

TMP: Temperature Alert

EMR: SOS Alert

TA: Tilt Alert⁸

27. BTYPE⁹ - Buzzer Configuration

Set or select buzzer type

Usage:

btype=<1/0>

1 – For Hooter type

0 – For Beeper type

28. RFCONF¹⁰ - RFID Reader Setting

To set RFID reader configuration

Usage:

rfconf=[timeout],[mode]

Timeout is in seconds for repeat card detection, set to 0 to disable and only alternate cards will be detected.
mode – Card display format, “decimal” (default) or “hex”.

29. BIOCONF¹¹ - Biometric Reader Setting

To set Biometric/Fingerprint reader configuration

Usage:

bioconf=[operation],[argument]

Supported Operations:

Load: To load biometric template from URL

bioconf=load,[index],[URL]

where index can be 1 to 200 (biometric ID)

⁷ This command is only available in devices with buzzer installed.

⁸ Only available if MEMS sensor installed

⁹ This command is only available in devices with buzzer installed.

¹⁰ This command is only available in devices with RFID installed.

¹¹ This command is only available in devices with Biometric installed.

Del: To delete template

Bioconf=del,[index]

Where index can be 1 to 200. To delete all templates at once use:

Bioconf=del,all

30. IBCONF¹² –iButton Reader Setting

This command is used for configuring iButton reader operation

Usage:

IBCONF=[Timeout],[Mode]

Timeout is in seconds for repeat iButton detection. Setting this value to 0 disable same ID detection back to back. Default value is 2 sec.

Mode can be sent to 0 or 1, 0 as Normal mode and 1 as reverse mode. In reverse mode ID is sent to server in reverse order.

31. IMCONF – Immobilizer Configuration

This command is used to configure advanced configuration for immobilizer. To automate immobilizer control based on RFID/Biometric/iButton. The individual configuration of control devices can be done with RFCONF/BIOCONF/IBCONF –iButton commands respectively.

Command Format:

IMCONF=<Operation>,<Type>,<arguments...>

Operation can be “set” or “get” to set or get the configuration value respectively.

Type defines the configuration on which operation to be performed and based on type the argument list is decided. Following are the types:

CTL: Control Configuration

This configuration is used to set the source of control for Immobilizer. The source can be as follows:

0 – No Source (Default)

1 – Any scanned ID from RFID/Biometric/iButton reader (No validation)

2 – RFID Reader

3 – Biometric Reader

4 – iButton Reader

Usage:

IMCONF=<set/get>,CTL,<source>,[<mode>]

Mode is optional argument and can be any of the following. The operation mode is only valid when source is not set to 0 (No Source).

Normal: This is default mode of operation. In this mode immobilizer is disabled when a valid ID is found or STARTV command is sent to the device. Immobilizer is enabled again after ignition is turned off.

¹² This command is only available in device with iButton Reader installed.

Toggle: In this mode, immobilizer will be enabled if it was disabled previously or disabled if enabled previously by showing a valid ID to the reader. Ignition has no effect when this mode is selected.

TOUT: Normal Mode Timeout

This command configures timeout value (in seconds) to auto turn-on immobilizer. Default value is 60 seconds. When Immobilizer is disabled via a control source and ignition is not turned on within configured timeout value, Immobilizer will get turned on again automatically.

Usage:

IMCONF=<set/get>,TOUT,<timeout>

Timeout is value in seconds, default value is 60 seconds and minimum value is 10 seconds.

RFID: RFID Card Configuration

This configuration set RFID card number to be used for immobilizer control.

Usage:

IMCONF=<set/get>,RFID,<index>,<RFID Card Number>

BIO: Biometric Fingerprint Configuration

This configuration set fingerprint ID to be used for immobilizer control.

Usage:

IMCONF=<set/get>,BIO,<index>,<Biometric ID>

Biometric fingerprint ID starts from 1.

IBTN: iButton Configuration

This configuration set iButton ID to be used for immobilizer control.

Usage:

IMCONF=<set/get>,IBTN,<index>,<iButton ID>

Index in all configurations can be from 1 to 10, which means maximum 10 cards/Biometric IDs/iButtons can be configured for control.

32. ADCTUNE – ADC Configuration

Set ADC approximation level.

This command can be used to smooth jumps in ADC data. The level of approximation can be adjusted by user from 0 (low or no approximation) to 10 (high approximation, will take more time to reflect changes).

Default value of approximation is 1.

Usage:

ADCTUNE=5

33. SIM – SIM Information

Get SIM ID of SIM card installed in the device. This command requires no parameter.

34. HEADER – HTTP Header Configuration

Set custom HTTP Header during data post. Used in case of JSON only.

User can add custom headers to be added to HTTP POST request sent by device to server. Most of the time custom headers are needed for device authentication or security. Header is set in a key: value pair in the POST request.

Usage:

To set key and value together:

HEADER=set,<keyname>,<keyvalue>

e.g. HEADER=set, Authorization, c29tZSB2YWx1ZSBhcyBhdXRoIGtleQ==

The above will add following to the POST Header

Authorization: c29tZSB2YWx1ZSBhcyBhdXRoIGtleQ==

To change only Key Name:

HEADER=key,<keyname>

To change only Key value:

HEADER=value,<keyvalue>

To delete key, value:

HEADER=del or HEADER=delete

35. FCFG – Fuel Sensor configuration

This command is used to configure fuel sensors supported by device.

Usage:

FCFG=[Operation],[Value...]

The operation is the sub-command to run for fuel configuration. Based on operation value and parameter count can be different. Operations are explained below

SRC: Sensor Type

Device supports 5 different types of sensors which are pre implemented in the firmware.

FCFG=SRC,<type>

Type is as follows:

- 1 – Analog Sensor
- 2 – Modbus type Sensor
- 3 – UART type sensor
- 4 – Omnicomm Digital sensor
- 5 – Ultrasonic Fuel sensor

PORT: Serial port configuration

Sensors which connects over serial/modbus require port parameters to be configured.

FCFG=PORT,[Baud],[Com parameters]

Baud is Baudrate of com port. Default is 9600

Com parameters are serial port communication parameters like databits, parity, stopbits etc. The parameters are given in following format.

[Databits][Parity][Stopbits]

Databits: 7 or 8

Parity: n – None, e – Even, o – Odd, m – Mark, s – Space

Stopbits: 1 or 2

Default value is 8n1 for 8 bits, no parity and 1 stopbit.

SID: Slave ID

For omniconm and modbus devices, slave id is required. Default value is 99.

FCFG=SID,<ID>

OFF: To set sensor offset value

FCFG=OFF,<value>

Value is the offset to sensor reading can be positive or negative. Default 0.

CTL: Fuel data post process control

To smooth out any data abnormalities caused by vehicle motion etc., the data is post-processed inside the device before it is sent to server. This setting enables or disable post processing of sensor data. Default is off.

FCFG=CTL,<0/1>

0 – Off, 1 – On

FIL: Post process level

This operation controls the level of post processing done on data. Higher the value, slower will be the response to change in data. Default level is 2.

FCFG=FIL,<level>

Level can be anywhere from 0 to 10.

RESET: Reset Settings

This operation will reset fuel sensor settings stored on device to default.

36. UMODE – Aux UART Operation Mode

This command is used to configure operation mode of Auxiliary UART port. Default mode is application mode.

Usage:

UMODE=<mode>

Mode can be 0 for Application mode (default), 1 for transparent mode.

For detail on transparent mode see **Transparent Mode for Auxiliary UART**.

37. TPCFG – Transparent mode configuration

This command is used to configure transparent mode settings.

Usage:

TPCFG=<operation>,<value,...>

Operation are sub commands explained further in this section and Value list is optional and depends on operation being executed. When value is not present device will return current settings.

CONN – Set Transparent server configuration

TPCFG=CONN,[IP],[PORT]

PORT – Set UART port configuration

TPCFG=PORT,[BAUD],[Com parameters],[DIRCTL]

BAUD: is Baudrate of UART port, Default Baudrate is 9600

Com Parameters: UART communication parameters same as: **PORT: Serial port configuration**
default value is 8n1.

DIRCTL: This parameter is optional to enable direction control if external device is using RS485 port.

38. GEO – Configure geofence

Device support only one geofence which can be either a circle or a rectangle.

Usage:

GEO=<event type>,<fence type>,<fence parameters>

Event Type:

0: Fence disabled, when fence is disabled all other parameters are ignored.

1: Fence Enabled and Event generated on geofence entry.

2: Fence Enabled and Event generated on geofence exit.

3: Fence Enabled and Event generated on geofence entry and exit.

Fence Type:

0: Fence type circular

1: Fence type rectangular

Fence Parameters: based on fence type fence parameter vary

For Circular Geofence

GEO=<event type>,0,<latitude>,<longitude>,<radius>,<name>

Latitude: Latitude of center of circular geofence

Longitude: Longitude of center of circular geofence

Radius: Fence radius in meters, this parameter is optional. Minimum and default value is 100m.

Name: User defined name for geofence, this can be any alphanumeric value not more than 31 characters.

For Rectangular Geofence:

GEO=<event type>,1,<latitude 1>,<longitude 1>,<latitude 2>,<longitude 2>,name

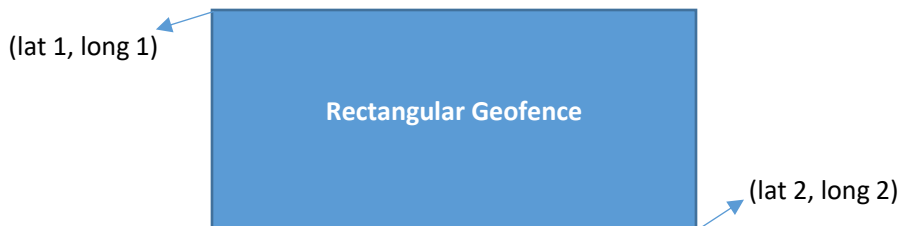
Latitude 1: Latitude of top left corner of geofence

Longitude 1: Longitude of top left corner of geofence

Latitude 2: Latitude of bottom right corner of geofence

Longitude 2: Longitude of bottom right corner of geofence

Name: User defined name for geofence, this can be any alphanumeric value not more than 31 characters.



39. IOCONF – Configure Input/Output Pins

VTS device can support up to 8 input/output lines however it is dependent on hardware feature available with the model. This command applies to IO available on the VTS device in use. IO lines are divided into two types:

- Input Only: Input lines are designed for input only operation and only input related parameters can be configured. Any output configuration for those pins is considered invalid and will not be applied. There are upto 4 Input only lines possible on a given VTS hardware.
- Input/Output Lines: These are general purpose IO lines which can be configured as input or output based on usage requirements. There can be upto 4 IO lines possible on a given VTS hardware.

Actual count of IOs vary based on hardware model.

Usage:

IOCONF=<ioname>,<direction>,<configuration>

Where ioname is ascii name given to IO lines for easy configuration. See **Table 19: IO Name and Specifications** for more information.

Table 19: IO Name and Specifications

IONAME	Description	Specification
IGN	Ignition Input	Input Only
BOX	Temper detect Input	Input Only
SOS	Emergency Switch	Input Only
TRIP	Trip Switch (when input)	Input Only
IO0	General Purpose IO 0	Input Only
IO1	General Purpose IO 1	Input (default)/Output
IO2	General Purpose IO 2	Input (default)/Output
IO3	General Purpose IO 3	Input (default)/Output
IO4	General Purpose IO 4	Input (default)/Output
IO5	General Purpose IO 5	Input (default)/Output
IO6	General Purpose IO 6	Input (default)/Output

Direction can be “in” for input or “out” for output line.

Configuration is based whether IO line is configured as input or output.

Input Pin Configuration

Usage:

IOCONF=<ioname>,in,<active_state>,<pull_select>,<debounce>,<trigger_type>

Ioname is same as mentioned in **Table 19: IO Name and Specifications**.

active_state: Active state of input signal. “high” for active high signal (default), “low” for active low signal.

pull_select: “pu” for pull-up, “pd” for pull-down, “none” for no pull.

debounce: Select debounce time for input trigger in milliseconds. Default 250ms.

trigger_type: “high” trigger on active, “low” trigger on inactive, “both” for any level (default).

Output Pin Configuration

IOCONF=<ioname>,out,<default_level>,<active_state>

Ioname is same as mentioned in **Table 19: IO Name and Specifications**.

default_level: default level of output line when no IO state is configured. “high” for default active state, “low” for default inactive state.

active_state: Select active state for output line, “high” for active high, “low” for active low.

NOTE: The active state defines the IO state for input and output lines e.g. For active low output, inactive level is high on the IO line and active level is low on IO line. Similarly, for Input line, active state for active low input line will be represented as 1 in iostatus inside data packet.

40. IOATTACH – Attach output line to event

This new feature lets you customize VTS device operation in lot of ways. By attaching an IO line configured as **output**, user can select how output line should behave on occurrence of an event.

Usage:

IOATTACH=<ioname>,<operation>,<configuration>

Ioname is same as that described in **Table 19: IO Name and Specifications**

Operation: Select operation type to perform on output line.

0 – For no operation, setting this will detach output line from event

1 – For Pulse operation, Output line will generate pulse based on selected function and configuration setting

2 – For Toggle operation, Output line will go active high when function event is active and low when inactive

3 – For Dual function operation, Output line will go active for one function and inactive on another

Configuration: Based on selected operation, configuration varies as follows.

Function:

Event type to attach the IO line with, this parameter is common in all configurations. See table below for list of available events

Table 20: Function Value for event type

Function Value	Type	Description
0	No Event	No function, disable attachment
1	Immobilizer Event	Runs on both Immobilizer on/off events
2	Immobilizer on Event	Runs on Immobilizer on event only
3	Immobilizer off Event	Runs on Immobilizer off event only
4	Motion Event	Runs on Motion start and stop events
5	Motion Start Event	Runs on motion start event only
6	Motion Stop Event	Runs on motion stop event only
7	Geofence Event	Runs when VTS enter/exit configured geofence
8	Geofence Enter Event	Runs only on geofence enter event
9	Geofence Exit Event	Runs only on geofence exit event

10	Trip Event	Runs on Trip start or end event
11	Trip Start Event	Runs only on Trip Start Event
12	Trip End Event	Runs only on Trip End Event
13	Overspeed Event	Runs when vehicle is over speeding
14	Low Battery Event	Runs when internal battery of device runs low
15	Tilt Event	Runs when tilt event is active
16	Temperature High Event	Runs on temperature high alarm
17	Temperature Low Event	Runs on Temperature low alarm
18	Humidity High Event	Runs on Humidity high alarm
19	Humidity Low Event	Runs on Humidity low alarm
20	RFID Event	Runs when new RFID card is detected
21	BIO Event	Runs when new Biometric fingerprint is detected
22	iButton Event	Runs when new iButton is detected
23	UHF Event	Runs when new set of UHF cards are detected
24	GPIO	GPIO transition event
25	GPIO High event	GPIO input high event
26	GPIO Low event	GPIO input low event
27	IMU Sensor event	IMU sensor motion detect event
28	IMU Motion active	IMU sensor motion detect active event
29	IMU Motion inactive	IMU sensor motion inactive event
30	Ignition Event	Ignition input transition event
31	Ignition On Event	Ignition/ACC line on event
32	Ignition Off Event	Ignition/ACC line off event
33	Box Temper	Box Temper alert event
34	Box Open Event	Box open event
35	Box Close Event	Box close event
36	Power Transit Event	Main power supply transition event alert
37	Power Connected Event	Main power supply connect is detected
38	Power Disconnect Event	Main power supply disconnect event
39	SOS Pressed	SOS button press event

Pulse Configuration

When pulse operation is selected, output line generates a pulse or pulses when event occurs.

Usage:

IOATTACH=<ioname>,1,<function>,<count>,<high_time>,<low_time>

Ioname and function is as described in previous section

Count: Number of pulses to generate on event occurrence.

High_time: Pulse high duration in milliseconds and multiples of 100. E.g. a value of 1000 means pulse with a high time of 1 sec.

Low_time: Pulse low duration in milliseconds and multiples of 100. E.g. a value of 1000 means pulse with a low time of 1 sec.

Toggle Configuration

There is no special configuration for toggle mode. The output line goes active high when event is active and goes active low when event is inactive.

Usage:

IOATTACH=<ioname>,2, <function>

Dual Mode Configuration

In this mode IO line can be attached to two events, one event set the line active high, and another sets it active low.

Usage:

IOATTACH=<ioname>,3, <function1>,<function2>

Function 1: Active high event

Function 2: Active low event

Function code as same as described in **Table 20**.

41. IOVAL – Set or get current level of GPIO lines

This command is only valid for GPIO lines only.

Usage:

IOVAL=<ioname>[,val]

Ioname is same as described in **Table 19: IO Name and Specifications**

Val is optional when setting level of output line. 1 means active high and 0 means active low level. The output state is decided by active state of the IO line configured using **IOCONF** – Configure Input/Output Pins command.

Usage for Pulse output:

IOVAL=<ioname>,pulse,<count>,<high_time>,<low_time>

Count: Number of pulses to generate

high_time, low_time: Pulse high and low duration respectively in milliseconds.

42. IMU – Configure IMU settings

This command is used to configure and calibrate Inertial Measurement Unit.

Usage:

IMU=<op>,<param>

Op: Operation or configuration to change.

Param: Value based on operation

TILT

Set tilt angle for event generation when vehicle is tilted

Usage:

IMU=TILT,<value>

Where value is angle in degrees. Default value is 45°

MTH

Set motion detection threshold.

Usage:

IMU=MTH,<value>

where value is time in seconds. Default value is 10s.

RST

Reset device mount position

Usage:

IMU=RST

DBG

Enable IMU sensor debug output

Usage:

IMU=DBG,<value>

Value is 1 to enable Yaw, Pitch, Roll output of IMU sensor

0 to disable debug output.

CAL

Calibrate IMU sensor

Usage:

IMU=CAL,<type>

Type set to 1 for sensor base calibration

Type set to 2 for zero calibration when vehicle is steady

i Contact our support team for help on how to calibrate IMU sensor.

Service Commands

1. STG – Get Device Settings

This command can be sent without SMS password, it is a service level command which can be shared with installation person without issue. This command is used to verify current device setting and get software information. This command is only allowed over SMS.

Response format:

U: <unitid>
IM: <IMEI>
SIG: <Signal strength>
CON: <Server connectivity status>
APN: <Current APN in use>
IP: <Configured server IP/URL>
PT: <Configured server port> (unused in JSON)
SW: <Software version installed>
HW: <Hardware version>

2. TEST – Device Test Parameters

This command is used to get device runtime information. Values of parameters vary from version to version, but response is kept same for consistency purpose. Unused fields will have a zero or no value. This command is only allowed over SMS and requires no password.

Response Format:

U: <Unit ID>
IM: <IMEI>
GP: <GPS Status>
MV: <Main power status>
BT: <Battery voltage>
GSM: <GSM Registration status>
<latitude, longitude>: <Current location>
SIG: <signal strength>
CON: <Connectivity status>
BX: <Device open/close>
IO: <GPI or GPO value>
AD1: <Analog 1 value>
IGN: <Ignition status>

3. WHERE – Locate Device

To get current device location information. Only applicable via SMS.

4. GPSINFO – Get Current GPS Parameters

This command can be used to get current device GPS parameters to verify GPS status etc. This command requires password if sent via SMS.

Device will respond in following format:

gps:<GPS Status>
Time:<UTC Time in HHMMSS>
Date:<Date in DDMMYY>
Fix:<A/V>
Sat:<Visible Satellites>
Speed:<Speed in km/h>
dir:<Direction degrees>
pdop:<PDOP value>
hdop:<HDOP Value>
Lat:<Current Latitude>
Long:<Current Longitude>
Alt:<Altitude in meters>
Odom:<Odometer in Km>

5. LOGLEVEL – Debug Log Setting

This command can only be used from Bluetooth terminal and is intended for device debugging on field. User can use Bluetooth application on phone to view on going device activity.

Usage:

LOGLEVEL=<debug level>

e.g. LOGLEVEL=1

Allowed debug levels:

1: To view Network related debug messages

24: To view packet data going on to server

Contact SiWi support (<https://help.siwi.in/>) if specific log level information or help is needed.

6. FRESET – Factory Reset

This command is used to reset device to factory settings. This command requires password to execute successfully, if sent via SMS.

NOTE: This command will remove all device configuration and format memory of the device. It is recommended to contact SIWI support before using this command.

Usage:

For SMS:

<SMS Password>,FRESET=confirm

For Server:

FRESET=confirm

7. FEATURE – Feature Configuration

This command is used to configure device optional features. Please contact support team for more information. This command requires password if sent via SMS.

Usage:

Feature – get currently enabled features

To set or enable a feature

Feature=<set/clear>,<feature name>

e.g. Feature=set,json

The above will enable data in JSON format over HTTP.

8. IDENTIFY – Get Device Identity

This command can be used to get information like, device cell number, IMEI, SIM ID, IMSI etc. This command can be sent from anywhere.

Usage:

identify=<reply number with +91>

e.g.: identify=+919876543210

Device will respond in following format:

<Device ID>,<IMEI>,<IMSI>,<SIM ID>,<Current Firmware Code>

e.g.: 1905090039297,869867036065573,404041042202747,89910473031822027477,CPLUS

9. FWCONFIG – Get firmware configuration

This command is used to get currently enabled firmware features.

10. FWCHECK – Check for new firmware update

To check for new firmware:

fwcheck=now

To disable auto firmware updates:

fwcheck=off

To enable auto firmware updates:

fwcheck=on

11. OTA – Firmware Update

This command is used for Over the air firmware upgrade. This command is allowed via SMS and TCP.

Password is mandatory for this command to work over SMS.

Usage:

ota=<firmware_filename>

Please contact SiWi support to get right firmware filename for the device.

⚠ Installing a wrong firmware might cause permanent damage to the device and will void warranty.